

MODEL QUESTION PAPER (REV-21)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

INTRODUCTION TO DATA MINING AND DATA ANALYTICS SET-I

(Maximum Marks : 75)

TIME : 3HRS

PART -A

Answer all questions in one word or one sentence

I.

1. What is Datamining?
2. List any two application of Datamining.
3. Define Supervised Learning.
4. In rule based classification, the learned model is represented as a set of ----- rules.
5. When will you say two events are independent.
6. List any two descriptive datamining models.
7. What is big data.
8. What was Hadoop named after?
9. _____ part of the MapReduce is responsible for processing one or more chunks of data and producing the output results.

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Differentiate datamining and data analytics.
2. Discuss about OLAP operations.
3. Explain the various analytical model requirements.
4. Describe confusion matrix.
5. Differentiate classification and prediction.
6. Differentiate K-means and K-medoid algorithm
7. Explain the disadvantage of Agglomerative Algorithm.
8. List down the characteristics of Big Data.
9. Discuss the application of big data analytics in credit risk modeling.
10. Define Map Reduce.

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Explain in detail the data analytics Life cycle.

OR

2. Explain the various Dimensionality reduction Techniques.
3. Find the best predictor using 1R algorithm.

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

OR

4. The values of x and their corresponding values of y are shown in the table below

X	0	1	2	3	4
Y	2	3	5	4	6

- a) Find the least square regression line $y = a x + b$.
 - b) Estimate the value of y when $x = 10$.
5. Suppose we have data on few individuals randomly surveyed. The data gives the responses towards interests to promotional offers made in the areas of Finance, Travel, Reading, and Health. Sex is the output attribute to be predicted. Apply Naïve Bayesian classification algorithm to classify the new instance (Finance=No, Travel=Yes, Reading=Yes, Health=No)

Finance	Travel	Reading	Health	Sex
Yes	No	Yes	No	Male
Yes	Yes	No	No	Male
No	Yes	Yes	Yes	Female
No	Yes	No	Yes	Male
Yes	Yes	Yes	Yes	Female
No	No	Yes	No	Female
Yes	No	No	No	Male
Yes	Yes	No	No	Male
No	No	No	Yes	Female
Yes	No	No	No	Male

OR

6. Explain the working of K Nearest Neighbour Classifier.
7. Consider the transaction data set given below. Set minimum support count as 2 and minimum confidence threshold as 70%. Find the frequent item set using Apriori Algorithm.

Transaction ID	List of Item_Ids
T100	I1,I2,I5
T200	I2,I4
T300	I2,I3
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I2,I3

OR

8. Explain K-means partitioning algorithm. What are its drawbacks.
9. Explain Agglomerative hierarchical clustering method.

OR

10. Use K-Medoid clustering algorithm to divide the following data into two clusters and compute the representative data points for the clusters.

X	2	3	3	4	6	6	7	7	8	7
Y	6	4	8	7	2	4	3	4	5	6

11. Illustrate the use of big data analytics in credit risk modeling.

OR

12. Discuss the architecture of HDFS and its features

(6x7=42)

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MODEL QUESTION PAPER (REV-21)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

INTRODUCTION TO DATA MINING AND DATA ANALYTICS -SET-2

(Maximum Marks : 75)

TIME : 3HRS

PART -A

Answer all questions in one word or one sentence

I.

1. What is Data Analytics?
2. List out the various OLAP operations?
3. Define unsupervised learning.
4. How to find the Euclidian distance between two points p and q.
5. What you meant by conditional probability.
6. State true or false- Agglomerative hierarchical clustering is a top down method .
7. List any two data analytics tools.
8. _____ function is responsible for consolidating the results produced by each of the Map() functions/tasks.
9. _____ can best be described as a programming model used to develop Hadoop-based applications that can process massive amounts of data.

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Explain the various data analytics types.
2. List out the various dimensionality reduction techniques.
3. Explain the benefits of data analytics.
4. Explain Naïve Bayes Algorithm.
5. Differentiate supervised and unsupervised learning.
6. What is a dendrogram?
7. Explain Support and Confidence.
8. Explain the 3 Vs of Big Data.
9. Describe the features of HDFS.
10. Explain the advantages of map reduce?

(8x3=24)

PART C

Answer all questions from the following. Each question carries 7 marks.

III

1. Briefly describe data warehousing and OLAP operations.

OR

2. Explain the analytic process model.
3. The job market is being studied in several neighbourhoods. Let X represents the total number of jobs in a given neighbourhood. Let Y represents the number of entry level jobs in the same neighbourhood. A sample of six neighbourhood gives the following information. For a neighbourhood with 40 jobs, how many are predicted to be entry level.

x	16	33	50	28	50	25
y	2	3	6	5	9	3

OR

4. Explain the working of K Nearest Neighbour Classifier.
5. Explain 1R algorithm in detail.

OR

6. Based on the following data determine the gender of a person having height 6 ft., weight 130 lbs. and foot size 8 in. (use Naive Bayes algorithm).

person	height (feet)	weight (lbs)	foot size (inches)
male	6.00	180	10
male	6.00	180	10
male	5.50	170	8
male	6.00	170	10
female	5.00	130	8
female	5.50	150	6
female	5.00	130	6
female	6.00	150	8

7. Explain Apriori Algorithm.

OR

8. Find the absolute support, relative support and confidence of the rule (bread => jam) in the following set of transactions
T1 {bread, butter},

T2{bread, jam, milk}

T3{Milk, curd},

T4{bread, jam}

9. Use K-Means clustering algorithm to divide the following data into two clusters and compute the representative data points for the clusters.

x_1	1	2	2	3	4	5
x_2	1	1	3	2	3	5

OR

10. Explain Agglomerative hierarchical clustering method.

(6x7=42)

11. Explain the various data analytics tools.

OR

12. Explain Big Data Analytics using Map Reduce and Apache Hadoop

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MODEL QUESTION PAPER (REV-21)

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

INTRODUCTION TO DATA MINING AND DATA ANALYTICS -SET-I- Answer key

(Maximum Marks : 75)

TIME : 3HRS

PART -A

Answer all questions in one word or one sentence

I.

1. The *process* of discovering interesting patterns and knowledge from *large* amounts of data.
2. Business intelligence, Search engines.
3. Algorithm “learns” from the training dataset by iteratively making predictions on the data and adjusting for the correct answer. .
4. IF-THEN rules
5. If the occurrence of one event does not have any effect on the occurrence of another, then the two events are said to be independent.
6. Clustering and association rule mining.
7. Big Data is a collection of data that is huge in volume, yet growing exponentially with time.
8. Cutting’s son’s toy elephant.
9. Map task

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Any three difference. 1marks each

Operational data base	Data warehouse
Data mining is a process of extracting useful information, patterns, and trends from raw data.	Data analysis is a method that can be used to investigate, analyze, and demonstrate data to find useful information.
The data mining output gives the data pattern.	The data analysis output is a verified hypothesis or insights based on the data.

It includes the intersection of databases, machine learning, and statistics.	It requires expertise in computer science, mathematics, statistics, AI.
It is also called KDD.	It is of various types - text analytics, predictive analysis, data mining, etc.

2. Any of these three operations. Roll-Up(Drill-up),Drill-Down ,Slice, Dice, Pivot(Rotate)
3. Business relevance, Statistical performance, Interpretable, Justifiability,...
4. A confusion matrix is used to describe the performance of a classification model (or “classifier”) on a set of test data for which the true values are known.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

5. Differentiate classification and prediction(Any 3 difference each 1mark).

Prediction	Classification
Prediction is about predicting a missing/unknown element(continuous value) of a dataset	Classification is about determining a (categorical) class (or label) for an element in a dataset
Eg. We can think of prediction as predicting the correct treatment for a particular disease for an individual person.	Eg. Whereas the grouping of patients based on their medical records can be considered classification.
The model used to predict the unknown value is called a predictor.	The model used to classify the unknown value is called a classifier.
The predictor is constructed from a training set and its accuracy refers to how well it can estimate the value of new data.	A classifier is also constructed from a training set composed of the records of databases and their corresponding class names

6. *K*-means attempts to minimize the total squared error, while *k*-medoids minimizes the sum of dissimilarities between points labeled to be in a cluster and a point designated as the center of that cluster. In contrast to the *k*-means algorithm, *k*-medoids chooses datapoints as centers.
7. Any two disadvantage(3marks)
8. Any three characteristics (3marks)

9. The role of Big Data Analytics in credit risk assessment is significantly vital. The use of this technology is the answer to almost all problems that fintech companies face in this area.

Big Data Analytics will ensure that fintech companies have all the information they need about specific customer groups as well as individuals. Once the problem of insufficient data is solved, proper analysis of the available information becomes essential to the process of credit risk assessment.

The ability of Big Data Analytics to assess and analyse high volumes of data in minimal time gives an edge to fintech companies utilising this technology.

Big Data Analytics not only provide all the relevant data, but they also analyse it accurately to provide considerably improved evaluations of credit risk.

10. MapReduce is a processing technique and a program model for distributed computing based on java. The MapReduce algorithm contains two important tasks, namely Map and Reduce. Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs). Secondly, reduce task, which takes the output from a map as an input and combines those data tuples into a smaller set of tuples. As the sequence of the name MapReduce implies, the reduce task is always performed after the map job.

(8x3=24)

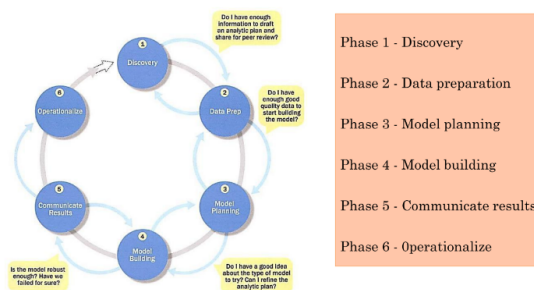
PART C

Answer all questions from the following. Each question carries 7 marks.

III

1.

Overview of Data Analytics Lifecycle



OR

2. Describe in detail any two dimensionality reduction techniques(each carries 3.5 marks)
3. The best predictor is

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3

IF Outlook = Sunny THEN PlayGolf = Yes
 IF Outlook = Overcast THEN PlayGolf = Yes
 IF Outlook = Rainy THEN PlayGolf = No

OR

4. a) $y = 0.9x + 2.2$ b) 11.2

5. Calculate the probability of sex being male for the given condition. Next check the probability of sex being female for the given condition. We classify the new instance to the highest probability we get. Suitable steps with correct prediction(7marks)

OR

6. Algorithm

Step-1: Select the number K of the neighbors

Step-2: Calculate the Euclidean distance of K number of neighbors

Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.

Step-4: Among these k neighbors, count the number of the data points in each category.

Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.

Step-6: Our model is ready.

7. Using apriori algorithm find the frequent item set.

OR

8. K-means partitioning algorithm(5marks) drawback(2marks)
 9. Explain the algorithm with an example.

OR

10. Using K-medoid algorithm find the data points of clusters

11. Write the use of big data analytics in credit risk modeling.

OR

12. Architecture(5marks) feature(2marks)

(6x7=42)

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MODEL QUESTION PAPER (REV-21)

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INTRODUCTION TO DATA MINING AND DATA ANALYTICS SET-2

(Maximum Marks : 75)

TIME : 3HRS

PART -A

Answer all questions in one word or one sentence

I.

1. Data analytics is the science of analyzing raw data to make conclusions about that information.
2. Drill down, Roll up, Slice, Dice
3. Uses machine learning algorithms to analyze and cluster unlabeled datasets
4. $d(p, q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}$.
5. The probability of the occurrence of an event A given that an event B has already occurred is called the conditional probability of A given B and is denoted by P(A/B)
6. False
7. Python, R, Tableau
8. Reduce
9. Map Reduce

(9x1=9)

PART B

Answer any eight questions from the following. Each question carries 3 marks.

II.

1. Any 2(Each carries 1.5mark) Predictive Analytics, Descriptive Analytics, Prescriptive Analytics, Diagnostic Analytics.
2. Describe any two such as Wavelet transform, PCA, etc.
3. Any three (Informed & improved decision making When big data joins forces with artificial intelligence, machine learning, and data mining, companies are better equipped to make accurate predictions. ..., More effective marketing What makes organizations stand out is the unique approach they choose to market their products. ...,Better & personalized customer experience ...,Streamline operations ...,Mitigate fraud ...

4.

OneR Algorithm

For each predictor,
 For each value of that predictor, make a rule as follows;
 Count how often each value of target (class) appears
 Find the most frequent class
 Make the rule assign that class to this value of the predictor
 Calculate the total error of the rules of each predictor
Choose the predictor with the smallest total error.

5. Any three difference

1. Supervised learning algorithms are trained using labeled data. Unsupervised learning algorithms are trained using unlabeled data.

2. Supervised learning model takes direct feedback to check if it is predicting correct output or not. Unsupervised learning model does not take any feedback.

3. Supervised learning model predicts the output. Unsupervised learning model finds the hidden patterns in data.

6. A *dendrogram* is a diagram that shows the hierarchical relationship between objects. It is most commonly created as an output from hierarchical clustering. The main use of a dendrogram is to work out the best way to allocate objects to clusters. Give an example.

7. The support of an itemset is the number of transactions in which the itemset appears, divided by the total number of transactions.

$\text{Support}(X) = (\text{Number of transactions containing } X) / (\text{Total number of transactions})$

Confidence is a measure of the likelihood that an itemset will appear if another itemset appears.

$\text{Confidence}(X \Rightarrow Y) = (\text{Number of transactions containing } X \text{ and } Y) / (\text{Number of transactions containing } X)$

8. Volume, Veracity, Variety

9. Cost-effective, Replication, Fault tolerance, Reliability,...

10. Scalability, Cost effective, Flexibility, security and authentication

1. $(8 \times 3 = 24)$

PART C

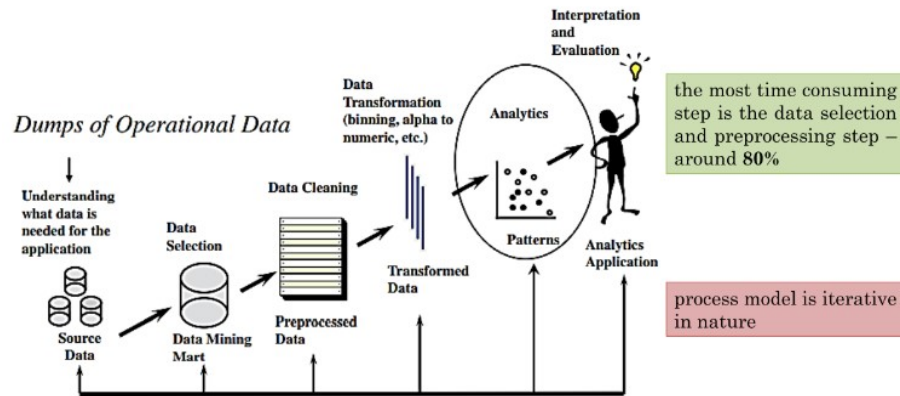
Answer all questions from the following. Each question carries 7 marks.

III

1. Describe data warehousing(2marks) and different OLAP operations(5marks).

OR

2.



3. Using linear regression method solve the problems Each carries 3.5 marks

OR

4. KNN classifier showing each steps.

5. 1R Algorithm in detail (7marks).

OR

6. Use Naïve Bayes classifier, determine the gender of the person.

7. Describe Apriori algorithm in detail.

OR

8. Support-3.5 marks, Confidence-3.5 marks

OR

9. Use K-means algorithm to compute the data points of clusters.

OR

10. Agglomerative hierarchical clustering method Explanation(7marks).

11. Explanation of any three tools may give full mark.

OR

12. Explain in detail